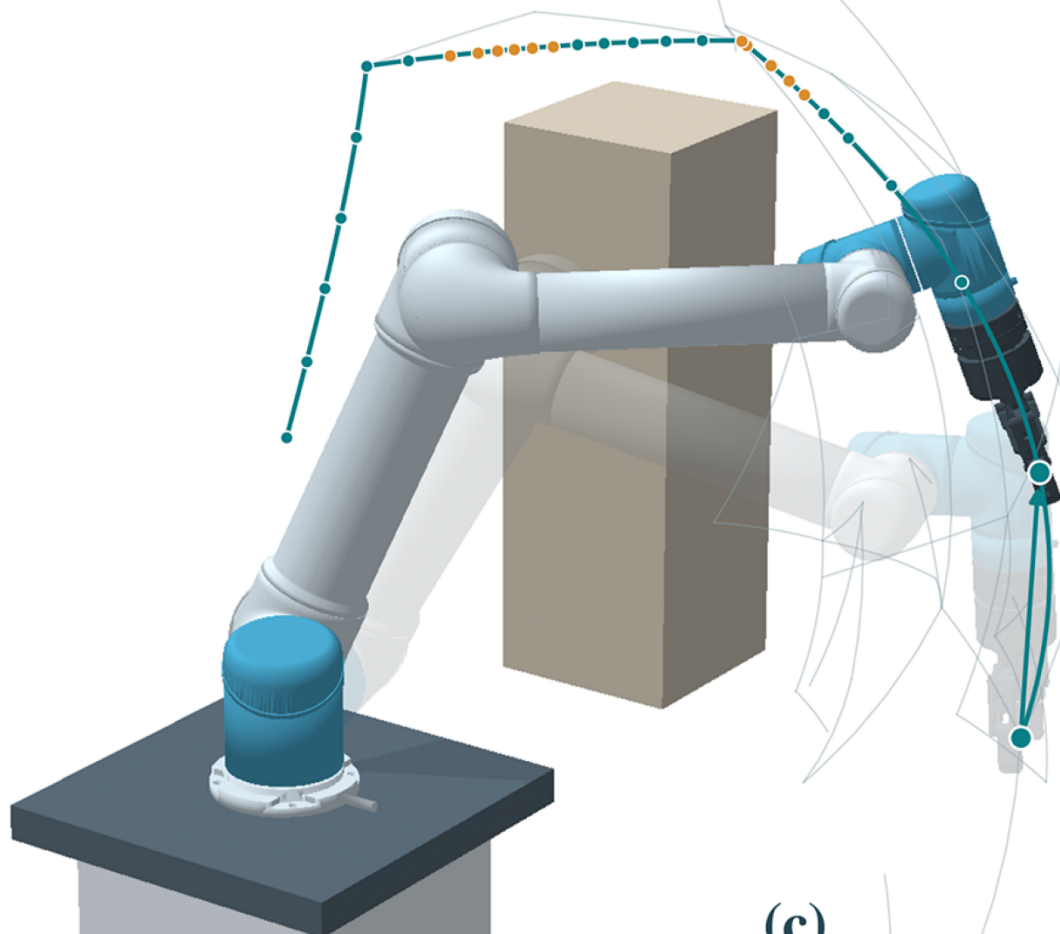


(a)

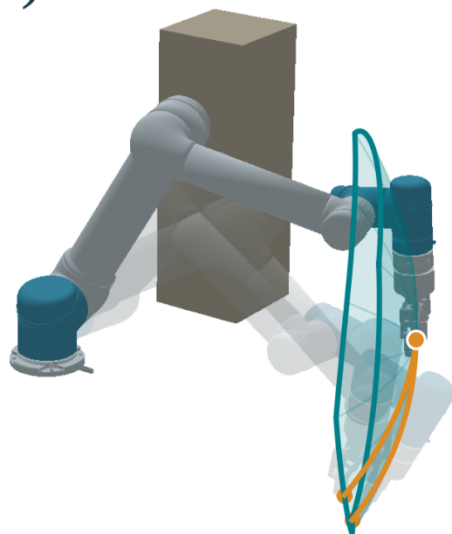
Smaller Steps
Near Obstacles



Larger Steps
Away Obstacles

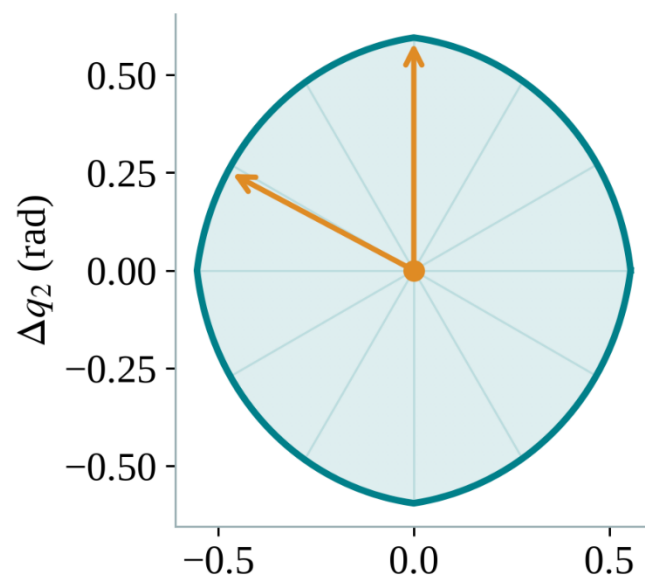
(c)

(b)



Kinematic image of the certificate

(c) Direction-dependent hold



$$\Delta q = \tau(u)u, \quad D_i(\tau; u) \leq h_i \quad \forall i$$